

## **Module 5 — Read 2**

**Inference for slope, mean-response CI, and prediction intervals**

# Module 5, Read 2: Inference in simple linear regression

**Project:** [Project 5 — Looking Ahead](#) · **Prerequisite:** [Read 1](#) and [Quiz A](#) · **Next:** [Quiz B](#), [demo lab](#), [exercise lab](#).

Complete Read 1 and Quiz A first. This reading adds **interval estimates** for the slope, the mean response, and individual predictions — and explains why prediction intervals are always wider.

## Learning outcomes

One section — use **Check yourself** before [Quiz B](#).

After this reading you should be able to:

### Section 1: Inference in simple linear regression

*Slope inference, mean-response CI, and prediction intervals*

- State the residual-normality assumption that supports interval inference in linear regression.
- Interpret a **95% confidence interval for the slope**  $\beta_1$  in context.
- Interpret a **confidence interval for the mean response** at a specific  $x_0$  value.
- Interpret a **prediction interval** for an individual future response at  $x_0$ .
- Explain why **prediction intervals** are wider than confidence intervals for the mean response.
- Use `confint()` and `predict(..., interval = "prediction")` in R to extract these intervals.

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### Section 1: Inference in simple linear regression

# Inference in simple linear regression

## Bridge from Read 1

You can already fit  $\hat{y} = b_0 + b_1x$  and make point predictions. Here we add **interval** estimates: a CI for the slope  $\beta_1$ , a CI for the **mean** response at  $x_0$ , and a **prediction interval** for one individual — and we explain why the PI is always wider.

## Inference in simple linear regression

If the linear model is appropriate and residuals are roughly normal with constant spread, we can do interval estimation:

- Confidence interval for the slope `beta1`.
- Confidence interval for the mean response  $E(Y \mid X = x_0)$ .
- Prediction interval for an individual future  $Y$  at  $x_0$ .

The prediction interval is wider than the mean-response confidence interval because it includes both uncertainty in the mean and individual-to-individual variation.

### In this section

When residuals are roughly normal, use `confint()` and `predict(..., interval=)` for interval inference.

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## Example 1 - Study hours and exam score (continued)

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

set.seed(3501)
n <- 160
exam_dat <- tibble(
  study_hours = runif(n, 1, 15),
  exam_score = 45 + 3.4 * study_hours + rnorm(n, 0, 8)
)
```

### Train/test split and fit

```
idx_train <- sample(seq_len(nrow(exam_dat)), size = 0.8 * nrow(exam_dat))
exam_train <- exam_dat[idx_train, ]
exam_test  <- exam_dat[-idx_train, ]
```

```
fit_exam <- lm(exam_score ~ study_hours, data = exam_train)
summary(fit_exam)
```

Call:

```
lm(formula = exam_score ~ study_hours, data = exam_train)
```

Residuals:

|  | Min      | 1Q      | Median  | 3Q     | Max     |
|--|----------|---------|---------|--------|---------|
|  | -16.0920 | -4.3112 | -0.8295 | 4.4390 | 21.3982 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 42.4805  | 1.3406     | 31.69   | <2e-16 *** |
| study_hours | 3.8436   | 0.1551     | 24.78   | <2e-16 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.664 on 126 degrees of freedom

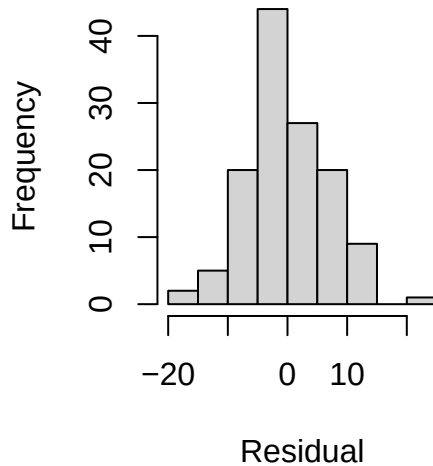
Multiple R-squared: 0.8298, Adjusted R-squared: 0.8284

F-statistic: 614.2 on 1 and 126 DF, p-value: < 2.2e-16

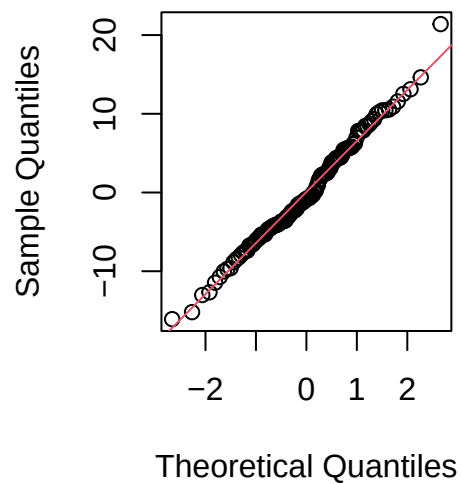
## Residual normality check (visual)

```
par(mfrow = c(1, 2))
hist(residuals(fit_exam), main = "Residual histogram", xlab = "Residual")
qqnorm(residuals(fit_exam)); qqline(residuals(fit_exam), col = 2)
```

**Residual histogram**



**Normal Q-Q Plot**



```
par(mfrow = c(1, 1))
```

## 1) CI for slope $\beta_1$

```
confint(fit_exam, "study_hours", level = 0.95)
```

```
          2.5 %    97.5 %  
study_hours 3.536713 4.150568
```

Interpretation pattern: we are 95% confident that each additional study hour changes mean exam score by a value in this interval.

## 2) CI for mean response at $x_0$

For  $x_0 = 10$  study hours:

```
x0 <- tibble(study_hours = 10)  
predict(fit_exam, newdata = x0, interval = "confidence", level = 0.95)
```

```
      fit      lwr      upr  
1 80.91687 79.56421 82.26953
```

Interpretation: 95% CI for the mean exam score among all students who study 10 hours.

## 3) Prediction interval for an individual response at $x_0$

```
predict(fit_exam, newdata = x0, interval = "prediction", level = 0.95)
```

```
      fit      lwr      upr  
1 80.91687 67.65902 94.17473
```

Interpretation: 95% interval for a single future student's score if they study 10 hours.

## Training vs test performance reminder

```
mse <- function(y, yhat) mean((y - yhat)^2)  
  
train_pred <- predict(fit_exam, newdata = exam_train)  
test_pred  <- predict(fit_exam, newdata = exam_test)  
  
train_mse <- mse(exam_train$exam_score, train_pred)  
test_mse  <- mse(exam_test$exam_score, test_pred)  
  
train_r2 <- 1 - sum((exam_train$exam_score - train_pred)^2) /  
  sum((exam_train$exam_score - mean(exam_train$exam_score))^2)  
test_r2  <- 1 - sum((exam_test$exam_score - test_pred)^2) /  
  sum((exam_test$exam_score - mean(exam_test$exam_score))^2)  
  
tibble(  
  set = c("train", "test"),  
  MSE = c(train_mse, test_mse),  
  R2 = c(train_r2, test_r2)  
)
```

```
# A tibble: 2 x 3  
  set      MSE      R2  
<chr> <dbl> <dbl>
```

```
1 train  43.7 0.830
2 test   66.3 0.767
```

Use test metrics as your main generalization report:

- Test MSE = typical squared prediction error on new cases.
- Test  $R^2$  = proportion of test-set variation explained by the linear model.

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## Example 2 - Fertilizer dose and plant height

```
set.seed(3502)
plant_dat <- tibble(
  dose = runif(95, 0, 12),
  height_cm = 18 + 2.2 * dose + rnorm(95, 0, 5.3)
)

idx <- sample(seq_len(nrow(plant_dat)), size = 0.8 * nrow(plant_dat))
plant_train <- plant_dat[idx, ]

fit_plant <- lm(height_cm ~ dose, data = plant_train)
summary(fit_plant)
```

Call:

```
lm(formula = height_cm ~ dose, data = plant_train)
```

Residuals:

|  | Min      | 1Q      | Median | 3Q     | Max    |
|--|----------|---------|--------|--------|--------|
|  | -10.7136 | -3.3558 | 0.5898 | 3.7855 | 8.9421 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 18.3691  | 1.3412     | 13.70   | <2e-16 *** |
| dose        | 2.1394   | 0.1706     | 12.54   | <2e-16 *** |
| ---         |          |            |         |            |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.961 on 74 degrees of freedom

Multiple R-squared: 0.68, Adjusted R-squared: 0.6756

F-statistic: 157.2 on 1 and 74 DF, p-value: < 2.2e-16

```
# CI for slope beta1
```

```
confint(fit_plant, "dose")
```

|      | 2.5 %    | 97.5 %   |
|------|----------|----------|
| dose | 1.799457 | 2.479396 |

```
# CI for mean response at x0 = 7
```

```
new_x <- tibble(dose = 7)
```

```
predict(fit_plant, newdata = new_x, interval = "confidence")
```

|   | fit      | lwr     | upr     |
|---|----------|---------|---------|
| 1 | 33.34505 | 32.2105 | 34.4796 |

```
# Prediction interval for an individual at x0 = 7
predict(fit_plant, newdata = new_x, interval = "prediction")
```

```
      fit      lwr      upr
1 33.34505 23.3956 43.2945
```

---

### Example 3 - Exercise minutes and resting heart rate

```
set.seed(3503)
heart_dat <- tibble(
  exercise_min = runif(110, 0, 60),
  resting_hr = 79 - 0.30 * exercise_min + rnorm(110, 0, 4.8)
)

idx <- sample(seq_len(nrow(heart_dat)), size = 0.8 * nrow(heart_dat))
heart_train <- heart_dat[idx, ]

fit_heart <- lm(resting_hr ~ exercise_min, data = heart_train)
summary(fit_heart)
```

Call:

```
lm(formula = resting_hr ~ exercise_min, data = heart_train)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-11.6123  -3.1631   0.3452   2.9383  14.2460
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  80.00362    1.17719   67.962 < 2e-16 ***
exercise_min -0.31807    0.03237  -9.827 1.02e-15 ***
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 4.854 on 86 degrees of freedom

Multiple R-squared: 0.5289, Adjusted R-squared: 0.5235

F-statistic: 96.57 on 1 and 86 DF, p-value: 1.023e-15

```
# CI for slope beta1
confint(fit_heart, "exercise_min")
```

```
      2.5 %      97.5 %
exercise_min -0.3824177 -0.2537277
```

```
# CI for mean resting HR at x0 = 30 min
new_x <- tibble(exercise_min = 30)
predict(fit_heart, newdata = new_x, interval = "confidence")
```

```
      fit      lwr      upr
1 70.46144 69.41853 71.50435
```

```
# Prediction interval for one person at x0 = 30 min
predict(fit_heart, newdata = new_x, interval = "prediction")
```

|   | fit      | lwr      | upr      |
|---|----------|----------|----------|
| 1 | 70.46144 | 60.75523 | 80.16766 |

---

## Key takeaways

- Slope CI answers: what plausible values for the linear effect **beta1** are supported by data?
  - Mean-response CI answers: where is the average outcome for all units at  $x_0$ ?
  - Prediction interval answers: where might one future individual outcome fall at  $x_0$ ?
  - Keep reporting test MSE and test  $R^2$  for predictive performance on new data. ::: {.callout-tip icon=false} ###  
Check yourself — Inference in simple linear regression
1. State the residual assumption needed before computing CIs and prediction intervals in regression.
  2. Output: `confint(fit, 'study_hours')` → [2.44, 3.68]. Interpret this 95% CI for the slope.
  3. At  $x_0 = 10$  hours, `predict(fit, interval = 'confidence')` gives [76.1, 80.5]. Interpret.
  4. At  $x_0 = 10$  hours, `predict(fit, interval = 'prediction')` gives [62.3, 94.3]. Interpret and compare to the CI above.
  5. In one sentence, explain why prediction intervals are always wider than confidence intervals for the mean response.
- :::

## Answers — Inference in simple linear regression

1. Residuals should be approximately **normally distributed** with **constant spread** (homoscedasticity) across all  $x$  values.
2. We are 95% confident that each additional hour of study changes the **mean** exam score by between 2.44 and 3.68 points.
3. We are 95% confident the **mean** exam score for all students who study 10 hours is between 76.1 and 80.5 points.
4. We predict with 95% confidence that **one individual** student studying 10 hours will score between 62.3 and 94.3. **Wider** than the mean-response CI because it adds individual-to-individual variation on top of uncertainty in the mean.
5. Predicting **one individual** involves both uncertainty about the mean (like the CI) **plus** the individual's random deviation from the mean.